

Solution Requirements

Date	1 October 2026
Team ID	SWTID-2026-1235
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Basic access to the web app; API key for Gemini is kept server-side in .env (user login is out of current scope).	Low
2	Authorization levels	Single role (Learner). Backend endpoints are the only component allowed to call the Gemini API.	Low
3	External interfaces	Integration with the Google Gemini API through the backend; REST communication between web interface and FastAPI.	High
4	Transaction processing	Each request is validated, converted to a prompt, sent to Gemini and the response returned as JSON.	High
5	Reporting	Results shown on screen as explanations, quizzes and summaries; no exportable reports in this version.	Medium
6	Business rules	Empty or oversized input is rejected; responses must follow the expected structure before display.	High
7	Compliance to laws or regulations	No personal data is collected; API keys are never committed to the repository; Google Gemini API terms are followed.	Medium
8	Other	Responsive interface and clear error messages when the AI service is unavailable.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Responses should appear quickly for typical queries.	Average response time < 10 seconds
2	Scalability	Stateless FastAPI service that can be containerised and scaled.	Supports concurrent requests limited only by Gemini API quota
3	Security & Data Privacy	API key stored in environment variables; input validated.	Key never exposed to the client; .env in .gitignore
4	Reliability & Availability	Graceful handling of API failures and timeouts.	Friendly error shown; no application crash
5	Usability & Accessibility	Simple, readable, mobile-friendly interface.	Usable on desktop and phone screens
6	Other	Maintainability – modular code, documented setup.	Project runs locally from README steps